

Decision Tree

Model	max_depth	min_samples_leaf	ccp_alpha	Train R ²	Test R ²
m1	10	1	0.0	1.000	0.9123
m2	10	5	0.0	0.9265	0.9112
m3	None	1	0.001	1.000	0.9123
m4	10	5	0.001	0.9265	0.9112

'''Models m2 and m4 show better generalization because the training and test R² values are close, indicating reduced overfitting. Although their test R² is slightly lower than the fully grown tree, they are more stable and less complex.'''

1. Base model:

Training R² = 1.0

Test R² = 0.917

Gap = 0.083

This means:

Tree memorized training data

High variance

Performance may drop badly on new unseen data

2. m2 / m4:

Lower training R²-0.926

test R²-0.911

Very small gap

More stable, robust, trustworthy..